POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2025
Programme: BE | Full Marks : 100
Course: Data Structure and Algorithms (New) | Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is Algorithm Analysis? Analyze the time complexity (best, and worst case) of the following code: 7

```
for(i = 0; i < n; i++)
for(j = 0; j < n; j++)
print(i + j);
```

b) Explain the algorithm design technique "Divide and Conquer" with a suitable example. 8

2. a) Explain the role of stack on conversion of infix to postfix expression. 8

OR

Convert following infix expression to postfix using stack.

A + B * (C ^ D - E) ^ (F + G * H) - I

b) What is a base case and recursive case in recursion? Explain how the problems are solved using recursion with a suitable example. 7

3. a) What is a circular queue? Write a C/C++ code to implement it using array. 8

b) Define a linked list. Write an algorithm to insert a node at the beginning of a singly linked list. 7

4. a) Explain the linked implementation of a queue in detail. 7

b) Explain the advantage of storing data in a hierarchical structure like in a balanced binary search tree. 8

OR

Construct a binary search tree (BST) from:40, 20, 10, 30, 60, 50, 70. Then delete node 20 and show the updated tree.

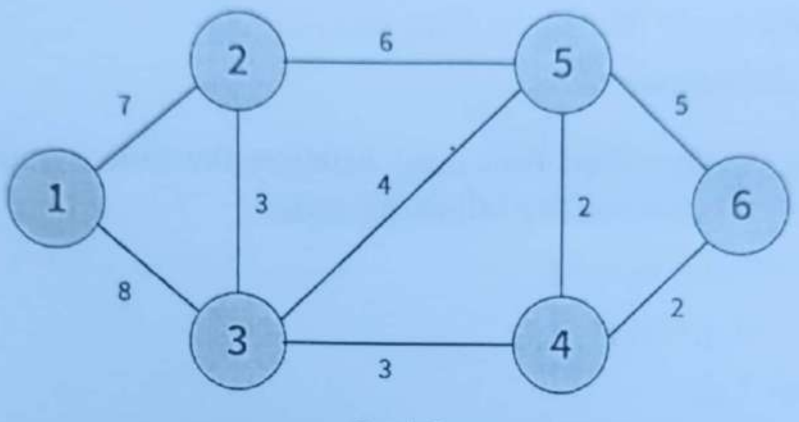
5. a) Construct an AVL-tree for the following data: 7

21, 26, 30,9,4,14,28,18,15,10,2,3,7

b) Differentiate between internal and external sorting. Sort the following numbers using the heap sort 20,70,10,20,50,30,25,18. 8

6. a) What is a hash system? Explain Linear Probing and Quadratic Probing with examples. 8

b) Find the minimum spanning tree of the following graph using Kruskal algorithm. 7



Graph G

7. Write short notes on: **(Any two)** 2×5

a) B tree and its need

b) Radix sort

c) Warshall's algorithm